Tejal Anavekar

CS graduate from USC with experience at Meta developing large-scale data pipelines and analytics platforms, complemented by full-stack engineering, data infrastructure, and AI/LLM-powered systems.

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Education

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | Graduated May 2026
- Relevant coursework: Analysis of Algorithms, Database management, Applied Natural Language Processing, Information Retrieval, Geospatial Information Management

Fr.C.Rodrigues Institute of Technology — Mumbai, India
Bachelor of Engineering in Information Technology | Graduated June 2024
- Relevant coursework: Operating Systems, Computer Networks, Artificial Intelligence, Big Data, Cloud Computing

Technical Skills

Languages: Python, C, C#, HTML, CSS, SQL, Javascript, Typescript
Engineering Practices: Data Pipelines, REST API Design, System Design, OOP, ETL/ELT
Frameworks: React, Node.js, Express.js, Next.js, ASP.NET Core, REST APIs, Minimal API, Flask
Libraries & AI Frameworks: LangChain, AI Agents, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy
Databases: PostgreSQL, MongoDB, Supabase, Firebase, SQL Server (T-SQL), Redis
Cloud & Tools: AWS (EC2, S3), GCP, Git, Docker, Linux/Unix, Shell Scripting

Experience

Data Engineer Intern — June 2025 – Aug 2025
Meta Platforms — *Burlingame, CA*
- Built and productionized 4 large-scale data pipelines on Meta's internal infrastructure, processing billions of records daily through distributed ingestion to drive real-time analytics and AI readiness dashboards across internal platform operations
- Developed Python-based data validation frameworks enforcing schema integrity, null-safety, and idempotency guarantees
- Optimized DAG-based data workflows through parallel execution with solutions adopted by 3+ cross-functional teams

Software Developer — Sept 2024 – May 2026
USC Institute for Creative Technologies — *Los Angeles, CA*
- Engineered full-stack internal platforms using C# ASP.NET Core Razor Pages, integrating Minimal API endpoints and dependency injection patterns for efficient and structured payload delivery across internal services
- Architected normalized SQL Server schemas, stored procedures, and EF Core data access layers, leveraging LINQ query optimization to accelerate dynamic reporting workflows and reduce Excel report generation time by 40%
- Deployed real-time dashboards to production, implementing end-to-end data validation and monitoring at live environment

AI Engineer — June 2022 – Apr 2023
Tata Institute of Fundamental Research — *Mumbai, India*
- Built CV pipelines integrating SIFT feature extraction, improving classification precision by 25% over CNN baselines
- Fine-tuned and deployed ResNet-50 and VGG variants, reducing inference latency by 30% in high-throughput workloads
- Scaled modular preprocessing pipelines like resolution normalization, histogram equalization, brightness calibration for robust, standardized input handling across heterogeneous datasets in production-grade ML pipelines

Projects

FixIt - AI Powered Debugging Platform | *React, Typescript, Node, Express, Supabase, Groq LLM* — June 2026 – Present
- Architected a debugging platform transforming software errors into interactive learning workflows and adaptive assessments
- Engineered schema-constrained LLM pipelines with Zod validation across 12+ structured fields, transforming natural-language debugging queries into syntax breakdowns, concept explanations, and knowledge-validation quizzes
- Implemented session-aware user state management and persistent sandbox execution using Supabase Auth and PostgreSQL

Budget Buddy | *MongoDB, Express, React, Node, LangChain, RAG, REST API* — Jan 2026 – Present
- Architected a modular full stack system with decoupled service and UI layers, improving scalability and maintainability
- Implemented secure identity and access management workflows using JWT, bcrypt for protected financial records
- Built RAG-powered LangChain pipelines for personalized financial intelligence, improving recommendation accuracy by 25%

Gene Semantic Bridge | *Pytorch, CVAE, BioBert, scRNA-seq, Generative Modeling* — Jan 2026 – May 2026
- Designed KA-CVAE (Conditional VAE) integrating BioBERT embeddings to model unseen CRISPR perturbations
- Engineered parameterized architecture with 128-dim latent space modeling across single and combinatorial perturbations
- Evaluated on Norman scRNA-seq OOD benchmarks, achieving R-squared value of 0.996, outperforming baseline models

GeoForecast - Weather App | *MongoDB, Express, React, Node, Flask, Swift, GCP* — Aug 2024 – Dec 2024
- Designed microservices-based system decoupling geolocation, weather, and user services into deployable modules
- Deployed containerized APIs on GCP Cloud with autoscaling, achieving 93% location accuracy, and +35% user retention
- Integrated geolocation and weather REST APIs with caching for low-latency real-time and daily forecast delivery